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Abstract

Organisations in Pakistan are faced with the problem of disparate data and costly alternative software solutions. Konvert is an AI-integrated business management software for time tracking and sales. It's designed to be distributed to organizations and handles an eight-tiered hierarchy, ranging from sales representatives all the way up to the CEO, providing a single source of data.

Automating repetitive task forecasts using AI with live tracking of routes using its motion detection capabilities. streamlines sales processes and reduces management overhead in a secure environment. Architecture is scalable from a small business to a large enterprise. It not only saves money from importing expensive software, but also helps in keeping business data locally which in turn helps in national business management.

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1 INTRODUCTION AND BACKGROUND

1.1 Project Motivation

The businessmen with good products have reached a digital dead end due to the fact that the sales management solutions which are available are either too costly or too difficult. This makes the business people spend more time on solving the technical issues rather than concentrating on sales. Using the foreign software for the purpose is also dangerous for the security of the data and the ownership rights. The problem which arises in the sales management is the lack of the local, secure, and easy-to-use solution where the business data can be stored.

1.2 Overview of the Proposed System

Konvert is an AI-driven sales management solution that aims to unite field work and executives, being the key automation solution for Pakistani businesses.

Going beyond the constraints of conventional management solutions, Konvert creates a state-of-the-art technical connection between field work on mobile devices and web-based platforms. Konvert is engineered to control and optimize an eight-layer organizational structure in order to make sure that information from the Sales Representatives at the street level is instantly available in the executives' dashboards. Integrating with the current web-based management systems, for example, Almashreqs infrastructure in case of Bristol, Konvert breaks the barriers of isolated data pools and creates a single source of truth.

The central component of the system is a forward-looking intelligence engine powered by machine learning and cutting-edge API that helps achieve ROI through live route tracking, automated tours, and AI-powered forecasting. Such predictive capabilities enable businesses to move away from reactive measures aimed at correcting their mistakes towards proactively expanding their operations and predicting future scalability and market trends. The system is scalable and suitable for use by enterprises of all sizes and automates sales-related processes while ensuring the security of national data.

1.3 Problem Description

Pakistani companies have a problem. They have to choose between a local solution and a global software product that is too expensive. Using foreign software causes a lot of issues. There are problems with the data security is not good. Companies do not have control over their operations.

Companies like Bristol are facing two issues. The local platform SanSFE is not user friendly. It does not work well with the main systems. This results in loss of data. On the hand global software products like Salesforce and HubSpot are too costly and complicated for Pakistani companies. Pakistani companies need something that works for them not something that's too expensive or too hard to use. The choice between SanSFE and software products like Salesforce and HubSpot is a difficult one, for Pakistani companies.

Lack of synchronization between applications in the field and in centralized data bases causes more than 20 per cent error in reporting. If it is not aligned, it causes inefficiency in the operation and causes strategic decisions to be taken and incorrect.

The main problems needing immediate solution are:

1. **Data Inconsistency:** When there is lack of 2-way synchronization between the mobile applications and conflicts occur between the data on the table and the data in the database, and duplicate work results.
2. **Operational Stagnation:** Route is not optimized locally and makes use of the AI, there are

inefficient processes and resource waste.

3. **Decision Making Challenges:** The lack of real-time visibility and predictive forecasting by managers up to the CEOs leads to reactive management.
4. **Data Privacy Issues:** Information about businesses hosted in the country on foreign servers puts businesses at a risk of being violated and at the same time under a threat of long-term security risks.
5. **Cost and Accessibility Challenges:** Foreign subscription and learning curve cost is a factor to keep local entrepreneurs and enterprises away from the adoption of advanced technology.

If they do not have a proper system, then the Pakistan companies will keep on being losing money to foreign companies as they have to work on inaccurate data.

1.4 Goals and Objectives

We want our research to result into the main automation system of the Pakistani business that would make the scattered field data work like a single engine towards development. These are some of our aims:

1. **Ensure Data Consistency:** Construct an integrated two-way system between field applications and Web databases to have one central source of truth across the organization.
2. **Improve Workflow Efficiency:** Lighten the administrative burden by automating administrative sales, reporting and stock monitoring processes to enable them to focus on growth.
3. **Improve Data Analysis and Forecasting:** Use Artificial Intelligence performance analysis and prediction to transform raw, field data into predictive data.
4. **Improve Operational Efficiency:** Package route optimization solutions and Tours planning services in Pakistan at world-class standard and cost savings vs foreign software; increase ROI.
5. **Safeguard National Data Sovereignty:** Create a safe local environment to manage your business and keep your sensitive information safely in Pakistan.
6. **Ensure Scalability:** Create an elastic architecture which allows companies to easily scale.

Our mission is to revolutionize the business process in Pakistan by converting the sales management from a tedious manual process to a proactive business process which is based on data. We are dedicated to providing technical solutions of high quality, user-friendly, and economical to encourage local industry growth and momentum without relying on foreign technologies.

2 DOMAIN ANALYSIS AND STAKEHOLDER IDENTIFICATION

2.1 Customer and User Categories

Konvert's customers are divided into two business categories and three user categories. Both have influenced all that comes after in the rest of this chapter.

2.1.1 Business Segments

Pharmaceutical and Distribution Organizations: Those firms operate an eight level chain of command and the actual issue is taken time field data can be dead on arrival for anyone who can do something about it. Konvert, on the other hand, allows everyone to share one environment; enabling executives to view what is currently happening in the field right now, rather than a week later, while reading about it.

Small and Medium Enterprises (SMEs): Most businesses in Pakistan are SMEs who typically have budget constraints for serious sales technology. It's local, evenly sized, and low enough in price it actually matters — and it automates so much of the routine work that they can grow without a new layer manned with administrators, just to keep up.

2.1.2 User Categories

Business Executives: Executives are concerned with the broader picture national sales trends, regional performance, AI projections for planning expansion. What they really want to safeguard is market share and ROI.

Sales Managers: These are the ones managing teams every working day. Automated reporting and real-time visibility lift some of the tedious work off their shoulders and help them identify which routes require improvement and determine the correct coaching based on numbers rather than intuition.

Field Sales Representatives: Reps produce most of the system's data, and paperwork is their main complaint. The mobile interface handles route tracking, tour planning and inventory, leaving more time for the sale.

2.2 Stakeholder Classification

Konvert will need a combination of internal and external partners, and everyone has their own agenda.

2.2.1 Internal Project Stakeholders

Table 2.1: Summary of Internal Project Stakeholders

Stakeholder	Role in System
Project Development Team (Asim, Obaid, Farheen)	Responsibilities include handling full-stack engineering, training AI models, and developing two-way synchronization logic.
Project Advisor (Hafiz Ahsan Arshad)	Suggests strategies and confirms the technical approach to ensure that the project remains true to the standards of professional engineering practice and academic requirements

2.2.2 External Project Stakeholders

Table 2.2: Summary of External Project Stakeholders

Stakeholder	Role in System
Bristol Mayer Biotech (Abdul Razzaq, Business Unit Head)	Main end user group and lead pilot partner. They promote the operational context and are the primary testing ground for the eight-tier hierarchical visibility.
Almashreqs Soft. (Bashir Ahmed, CEO)	Maintains the web database in the center. They secure the API endpoints and maintain the Web-to-mobile data channel's operation.
Local SME Sector	Secondary group who validates the market, provides feedback about the accessibility and low cost of the system to non enterprise businesses.

2.3 Impact on Businesses and Society

Konvert benefits not only the companies that use it but also their employees and the local economy.

Impact on Society: By automating routine tasks such as data entry and reporting, field employees can spend more time on sales and customer interactions. This improves productivity and helps employees focus on more important work. The system is designed to be easy to use, so employees can learn it quickly with minimal training.

Economic Benefits: Konvert is a locally developed solution. This reduces the need for businesses to purchase expensive foreign CRM and ERP systems. As a result, more money stays within the country and local businesses receive better support. It also allows small and medium-sized enterprises (SMEs) to access modern business tools at an affordable cost.

Organizational Benefits: Within an organization, Konvert helps managers identify problems early by providing timely information and reports. Better communication between different levels of management improves coordination and supports better decision-making.

2.4 External Systems and Integrations

Konvert uses several external services to support its operations.

1. **Web Management System (Almashreqs):** The existing Almashreq web system is used as the main source of organizational data. Konvert connects to this system through APIs to exchange information between mobile and web platforms.
2. **Location and Navigation Services (Google Maps API):** The Google Maps Platform is used for location tracking, route planning, and distance calculations. These features help monitor field activities and improve tour planning.
3. **Machine Learning Frameworks:** Machine learning tools such as Scikit-learn and TensorFlow are used to analyze historical data and generate performance predictions.
4. **Google Firebase Services:** Firebase provides services such as real-time data synchronization, push notifications, and user authentication. These services help keep the application secure and responsive.
5. **User Authentication Services:** Google Authenticator is used to provide Multi-Factor Authentication (MFA), which adds an extra layer of security to user accounts.

2.5 Supporting Reference Documents

2.5.1 Review of Related Projects

To figure out where Konvert actually stands, we looked at what's already out there, regionally and globally.

SanSFE (Regional Sales Management Platform): SanSFE is commonly used in pharmaceutical distribution. It supports field reporting but has limited real-time synchronization between mobile and web systems.

Salesforce and HubSpot (Business Management Platforms): International solutions such as Salesforce and SAP offer many features, but they are often expensive for local SMEs and usually require additional customization.

Local ERP Solutions: Many local organizations still use older ERP systems. These systems are mainly designed for record keeping and often lack mobile support, real-time tracking, route planning, and predictive analytics. Konvert aims to provide these features in a single platform.

2.5.2 Competitive Feature Comparison

Table 2.3: Comparative Analysis of Platform Features

Feature	SanSFE (Regional)	Global CRM (Salesforce)	Konvert (Proposed)
Data Sync	Manual reconciliation	High-speed cloud sync	Localized two-way flow
Mobile UI	Outdated legacy design	Modern but complex	Simplified field-force UX
Reporting	Static PDF logs	Advanced BI dashboards	Automated local metrics
Cost	Moderate local	High dollar-based	Affordable for local SME
Visibility	Immediate superior only	Customizable/Complex	Native 8-tier hierarchy
Intelligence	Record-based only	High-cost AI add-ons	Built-in native AI engine
Offline Work	Poor/Unstable	Basic/Cloud-dependent	Strong offline-first engine
Onboarding	Lengthy setup	Long consultant cycles	Agile rapid deployment
Support	Limited regional	Remote/Indirect	Direct local engineering

2.5.3 Reference Documents Overview

Technical Standards and Quality Framework:

- ISO/IEC 25010:2011 – System and software engineering quality models.
- ISO/IEC/IEEE 12207:2017 – Software life cycle processes.
- IEEE Std 830-1998 – IEEE Recommended Practices for Software Requirements Specifications (SRS).

- NIST Special Publication 800-63B – Digital Identity Guidelines.
- Human-centered design for interactive systems.
- IEEE 1012 – Standard for System, Software, and Hardware Verification and Validation (V&V).

Technical specifications and implementation:

- The specification for W3C Geolocation API.
- Software Engineering Guide to the Software Engineering Body of Knowledge (SWEBOK Guide V3.0).
- Fielding, Richard Twyman. (2000) – Architectural Styles and the Design of Network-based Software Architectures (REST).
- Google Maps Platform Technical Reference.
- Firebase RTL Architecture Patterns.

Research and Data Sovereignty:

- Hochreiter, S., & Schmidhuber, J. (1997) – Long Short-Term Memory (LSTM).
- General Data Privacy Principles (GDPR/APPI) – Legal frameworks ensuring data sovereignty, residency, and privacy.

3 SYSTEM REQUIREMENTS ANALYSIS

3.1 Core System Requirements

3.1.1 Detailed Functional Requirements

The functional requirements listed below outline what Konvert needs to do in order to assist with the day-to-day sales process, starting from logon until end-of-day reporting.

The FR Module 1 focuses on initializing, securing and accessing the system. The module 1 of FR is About how to initialise, secure and access the system.

- FR-1.1 Security and stability checks, as well as checks for updates, have to be performed by the app during the sequence of the splash screen before access is granted.
- FR-1.2 **Error Handling and Recovery:** Self healing should be provided in the system when an error It happens at run time (cache clear, database re-sync, reinstall).
- FR-1.3 **Domain and Hierarchy Onboarding:** FS Users need to have access to a specific company domain (type Bristol) via API keys and download the appropriate 8 tier hierarchy.
- FR-1.4 **Multi-Factor Authentication (MFA):** secured by a password, plus a randomly generated 6-digit code in Google Authenticator. Almashreqs' database hash.
- FR-1.5 **Feature Locking Mechanism:** It must be able to access or reclaim features according to the role of the user in the hierarchy retrieved.

FR Module 2: Data Synchronization and Persistence

- FR-2.1 **Master Sync Protocol:** On login the system is to perform a complete sync with the Almashreq cloud computer system. Getting CRM, and only that UserID's data.
- FR-2.2 **Bi-Directional Real-Time Sync:** Sync "real-time" in both directions: Field orders, logs, should sync immediately with the Almashreq database, and updates from the web should reach the mobile app immediately.
- FR-2.3 **Offline-First Architecture:** The app must continue to be usable even offline and queue the transactions locally which will be uploaded automatically when it goes back online.

FR Module 3: Intelligent Dashboard and Analytics

- FR-3.1 **AI Performance Prediction:** User's dashboard should display an AI prediction of user's chances for success for the upcoming week, using previous behaviour.
- FR-3.2 **Target Visualization:** Daily and weekly graphs must report on the progress towards tours currently being executed.
- FR-3.3 **Automated Notification System:** A background [photo checker] shall retrieve and display the top priority alerts [warning, updates, approvals] at a separate notification center.
- FR-3.4 **Advanced Reporting Module:** Should remain able to generate performance reports in granular format (daily, weekly and categorical).

FR Module 4: Tour Planning and Roadmap

This is FR Module 4: Tour Planning and Roadmap.

- FR-4.1 **Automated Tour Plan Creation:** Users need to be able to make new tour plans by selecting dates, routes, and expense budgets. This unit contains no required assessment.
- FR-4.2 **AI Enhancement (“Enhance Button”):** The system should have an AI helper class, which pre-optimizes the tour plan before submission.
- FR-4.3 **Management Approval Workflow:** To make sure tour plans in FR-4.3 are approved before departing, they will be locked and cannot be changed until the management tier has approved a digital form of the plan.

FR Module 5: Fieldwork and Geospatial Operations

Activities / Resources for FR Module 5: Fieldwork and Geospatial Operations

- FR-5.1 **Integrated Route Navigation:** The fieldwork interface should have a Google Maps display to enable users to view the direction of the next team member’s location without exiting the app.
- FR-5.2 **Geo-Fencing Triggers:** Product Selection Buttons should remain disabled until when player’s GPS location puts them within a set radius of the customer.
- FR-5.3 **Quick-Update Mechanism:** Configure the fieldwork page to provide localized data for ‘quick’ retrieval from the server, without causing a Master Sync.

FR Module 6: Order Management and Booking

- FR-6.1 **Dynamic Product Selection:** allow the users to choose products from the OTC and Regular product categories, with the discount and bonuses as determined by the current logic.
- FR-6.2 **Daily Booking Summary:** Users should be able to view and edit/confirm daily bookings before final submission.
- FR-6.3 **End-of-Day Upload:** The system needs to process all daily bookings, upload them to the Almashreqs cloud CRM for administrative approval, and compile them into a single file to be uploaded to the cloud CRM at the end of the day.

3.1.2 Detailed Non-Functional Requirements

The quality requirements below are performance expectations that the system must achieve as well as design constraints and quality specifications that the system must adhere to.

NFR Module 1: Performance and Efficiency

- NFR-1.1 **Rendering and Sync Speed:** The Flutter UI should be able to navigate at least 15 FPS and the first Master Sync should complete within 15 seconds using a 4G connection.
- NFR-1.2 **Battery Optimization:** If there are services running in the background, such as a 5-minute polling against the GPS, other AI data collection, those services should consume no more than 15% of the device battery during an 8 hour shift for NFR-1.2 Battery Optimization.

NFR Module 2: Reliability and Availability

- NFR-2.1 **Offline Resilience:** No network failures: The system should minimise the amount of data loss from a network failure by using robust local caching before the system tries to sync with Firebase.
- NFR-2.2 **High Availability:** Executive monitoring and data flow have to be uninterrupted which requires NFR-2.2 High Availability, meaning the cloud infrastructure and Almashreqs API bridges must ensure 95% uptime SLA for the cloud.

Introducing the NFR Module 3 on Security and Compliance.

NFR-3.1 Encryption Standards: All data while in transit should be encrypted through TLS 1.3 and all data on the local cache should be encrypted through AES-256.

NFR-3.2 Breach Management: The system is supposed to auto-expire the idle sessions, wipe local data on the instant and inform the admins when it senses device security breach or root access.

NFR Module 4: Scalability and Maintainability

Konvert Functional Specifications, v1.0 8

NFR-4.1 Concurrent Architecture: A SQL structure and API should be able to support 50+ active field agents at the same time without causing write-lock bottle-necks.

NFR-4.2 Application Updates: Every tier must remain on a compliant software version after forced mandatory updates occur, as well as after additional optional delayed updates (days 1-2) occur. Forced mandatory application updates as well as additional optional delayed updates (days 1-2) are both supported by the system to ensure all tiers remain at a compliant application software version.

This concludes the forum for NFR Module 5: Usability and Compatibility.

NFR-5.1 Hardware Compatibility: The Application needs to perform well on less capable mobile devices. As these are given to most of the regional field representatives, at least 4Gb of RAM should be specified.

NFR-5.2 Human-Centered UX: Based on ISO9241-210, there are field workflows with core elements that no more than three clicks are needed to get at or three taps on touch screens to avoid friction between the person and the field machine.

3.2 Identified System Actors

Konvert touches the following stakeholders:

- **Executive Leadership (CEO / Business Unit Head):** Top-tier decision makers who need national-level information to make a decision. Level analytics and AI forecasting.
- **Tactical Managers (National / Regional / Zonal / Area):** Manager: A manager assigned to the middle tier of a team of tactical managers, National / Regional / Zonal / Area, who overlooks the team's efficiency. Manager: Oversees the efficiency of a team of tactical managers (National / Regional / Zonal / Area). For levels 2-7, approving tour plans and monitoring the performance.
- **Operations Support (Command Center Operatives):** Backend personnel who manage system alerts, data reconciliation, technical support.
- **Field Sales Representatives (Tier 8):** Front line employees conducting daily visits, making orders, generating field data.
- **System Administrator:** Configures systems, manages users, policies and creates security policy in the domain system updates

3.3 System Users Requirements

Table 3.1 outlines a broad set of examples for each of the systems. Table 3.1 lists a comprehensive catalog of System Use Cases.

Table 3.1: Comprehensive Catalog of System Use Cases

UCID	Use Case Name	Brief Description
UC-1	To start up and ensure system security.To boot-up and check system security.	Installs security, stability and update. It will check on launch of the app before granting access.
UC-2	User Authentication and Domain Setup	Supports integration with the domain of the company using API keys and retrieves the 8 layer hierarchy of it.
UC-3	Multi-Factor Authentication	Checks google authenticator tokens to the central database hash.
UC-4	Master Data Synchronization	Syncs with Almashreqs cloud CRM bi-directionally and on login.
UC-5	Offline Data Management	To support off-line data management, UC-5 Offline data management queues transactions data locally, while the network is severed. Just reverts back to loss and auto upload when reconnected.
UC-6	AI Performance Forecasting	The weekly success probability is established using historical patterns and is displayed.
UC-7	Target Visualization & Reporting	Provides live graphs of the daily and weekly target achievement versus tour plan.
UC-8	Notification Management	collects and presents “prioritized alerts” using the: notification center.
UC-9	Tour Plan Creation	Create weekly tour plans from UC-9 Tour plan creation. Routes, dates and budgets.
UC-10	Improve the AI tour plan in UCC-10.Enhance AI tour plan in UCC-10.	Optimize the route for efficiency through the use of AI-helper class and Google Map.
UC-11	Tour Plan Approval Workflow	Locks will submit plans until digital signals get approved from management.
UC-12	Route Navigation and Geo Fencing.	Includes Google maps for navigation and only displays product Buttons inside of a radius of the customer.
UC-13	Customer check-in, order booking.	Manages product selection, discount logic, voice remarks and photo proof capture.
UC-14	End of day data upload	collects all records of daily bookings and securely uploads them to Almashreqs CRM.
UC-15	System Administration and User Management	Participants are taken to the UC-15 system administration and user management. Manages domain configuration, role assignments, security policies, and forced updates.

3.4 Visual Use Case Diagrams

The options provided across the eight different roles in the hierarchy of functional abilities are difficult to present in one use case diagram without it getting too big, particularly when printed on A4 paper, so as a result, the Konvert options are split between use case diagrams at different layers. All the four functional modules thus make up four sub-diagram use cases of the use case model.

The four modules are:

- Only a user authorizes the distribution of a key bundle.
- Data synchronisation and Off-line management module
- Tour planning and field operations.
- Dashboard, analytics, and reporting module

The sub-diagrams are explained below, as well as how they relate to each other and to the actors.

3.4.1 Module 1: Authentication and Security

This module manages secure access to Konvert, including checking and managing the device's integrity, authenticating users, and enforcing the security policy. The domain-based authentication flow requires field sales representatives and tactical managers to log into the platform, while the system administrator is responsible for platform security and account management.

The flow begins with system startup and security check, verifies integrity and version compliance for applications, checks for root access, and more. Using user authentication and domain Setup, users log on from there. Turning on Multi-factor authentication brings as an extra step to that flow.

This includes activation, deactivation and monitoring of user accounts on every level under system administration and user management.

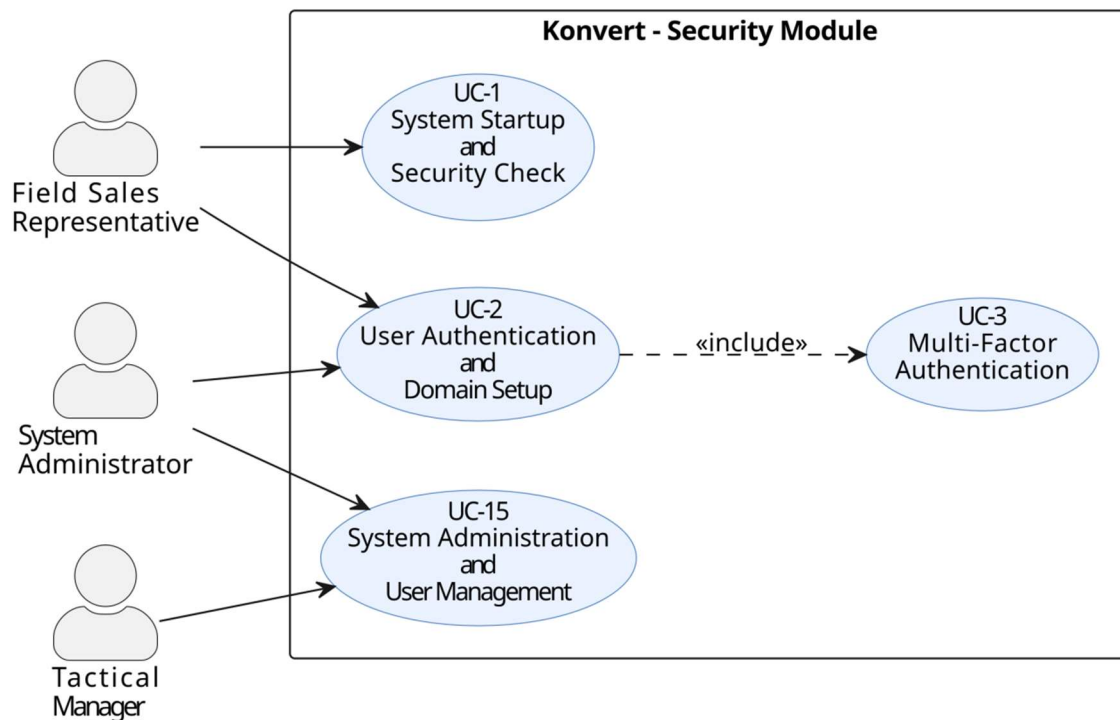


Figure 3.1: Visual use case model: Authentication Security

Figure 3.1 Description:

The diagram depicts the actor System Administrator connected to:

- The first driving stage is the System Startup and Security Check feature, available on the PlugSnell software (UC-1).
- Course UC-15: System Administration and User Management covers system administration and user management.

The Field Sales Representative and Tactical Manager actors are linked to:

- UC-3 is related to UC-2 by a «include» relationship. All use cases are within a system border marked with Konvert – Security Module.

3.4.2 Module 2 Data Synchronization and Tracking Offline.

This module copies operational data from mobile device to Almasreqs cloud service or vice versa, regardless of whether it's connected or not at any given time.

Master Data Synchronization loads the pending transactions offline and fetches the latest Customer, Product data, Target and Tour Plan. Once connectivity is restored, synced data is stored offline, synced data is handled by sync queues, and conflict resolution is managed offline.

Operations Support keeps an eye on sync status and chases down failed transfers. The Almasreqs External System is used as the back-end service and the sync relies on it.

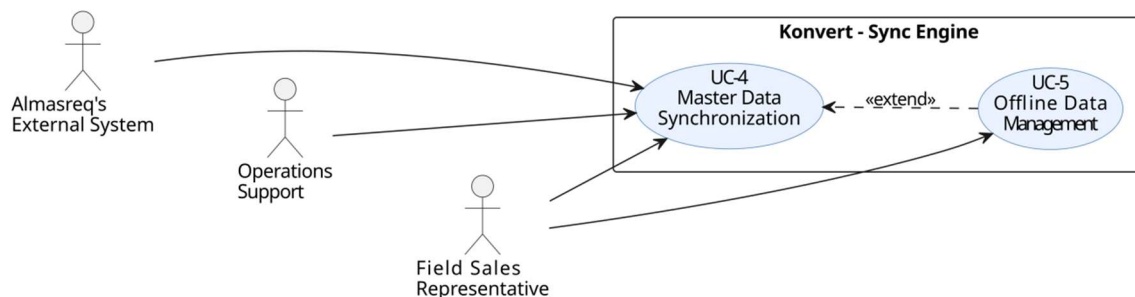


Figure 3.2: The visual-use case model of data synchronization is given as Figure 3.2.

Figure 3.2 Description:

There are the following actor interactions in the diagram.

- Field Sales Representative is translated as UC-4 Master Data Synchronization.
- Field Sales Representative to become UC-5 Offline Data Management
- → UC-4 Master Data Synchronization: synchronizing information across the enterprise or with specific applications.
- A custom external system connection via API from Almasreqs to UC-4.

If the condition network unavailable exists, then UC-4 is an extension of UC-5. All use cases are within the inner boundary "Konvert – Sync Engine".

3.4.3 Module 3: Tour Planning and Field Operations

This module contains the main tasks that a field rep is supposed to do: Editing a tour plan, optimizing the tour, navigating to each stop, checking in with a geo-fence, and recording the tour order.

Tour Plan Creation allows tour reps to create a tour plan for a given week. If needed, they can optionally proceed with AI Tour plan enhancement, where the route will be optimized. If approved, then that plan becomes the basis for the day's field activity.

Route Navigation and Geo-Fencing is assured that a rep is at a customers location before the order booking screen appears. After the geo-fence check, Customer Check-in / Order Booking is available. Tour Plans are reviewed and approved by Tactical Managers.

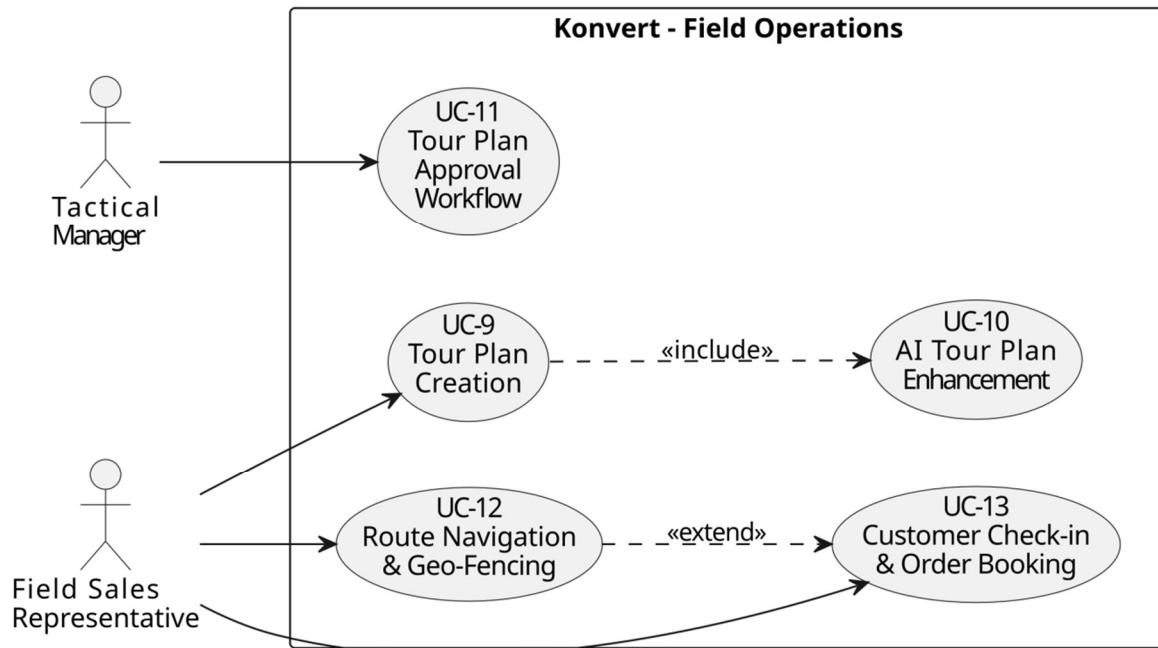


Figure 3.3: Visual Use Case Model, Tour Planning & Operations shown in Fig 3.3.

Figure 3.3 Description:

There are the following actor interactions in the diagram.

- Converted by the Field Sales Representative to UC-9 Tour Plan Creation.
- Field Sales Representative - UC-12 Route Navigation and Geo-Fencing
- Check-in/Booking Order Representative - Field Sales Representative
- Tactical Manager as a User type should allocate a game type with the User Centered–11 Game Plan Approval workflow.

UC-10 AI Tour Plan Enhancement is integrated into UC_9 when user clicks enhance. UC-12 enters GPS into geo-fence and extends UC-13. All Use Cases are within the system boundary labelled Konvert – Field Operations.

3.4.4 Module 4: Intelligence Dashboard and Reporting

This module provides sales performance figures, target tracking, alerts and forecasts for management. AI Performance Forecasting utilizes machine learning to analyze past data and predict future outcomes, that's what it does. Target Visualization and Reporting provides graphical dashboards, KPI tracking and

side-by-side performance comparisons. Notification Management sends notifications, reminders and operational information to the org.

Field Sales Representatives review their own targets and numbers on the reporting screens, and tactical managers and Executive leadership have a big picture view of the organization.

Flip Method SMART Functional Specifications v1.0 13

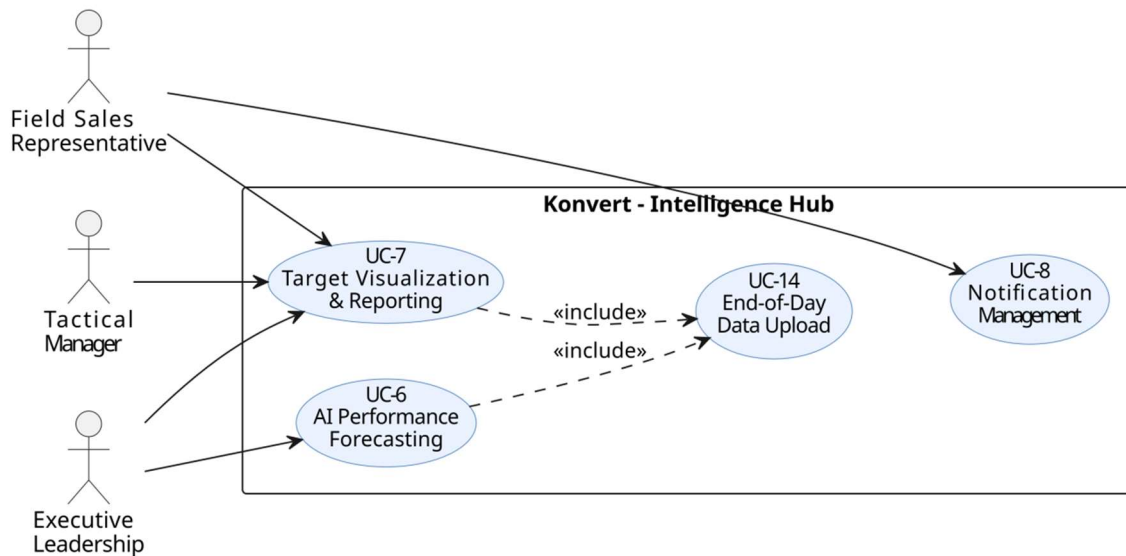


Figure 3.4: Figure 3.4: VUCM for Dashboard & Intelligence Analytics

Figure 3.4 Description:

There are the following actor interactions in the diagram.

- The AI viewpoint: Executive Leadership UC-6 AI performance forecasting.
- UC-7 will be used to target Visualization and Reporting to develop the DEA.
- The tactical manager application is replaced by the UC-8 Notification Management application.

UC-14 End of day data upload is required for both UC-6 and UC-7 through include relationships. All use cases are within the Konvert intelligence hub system boundary.

All four diagrams combined represent the relationship of those four elements security, sync, field operations, reporting across Konvert.

3.5 In-Depth Use Case Narratives

This course introduces users to the fundamentals of user authentication and domain setup.

3.5.1 UC-2: User Authentication and Domain Setup

Table 3.2: Detailed Flow for UC-2: User Authentication and Domain Setup

Use Case ID:	UC-2
Use Case Name:	As the submenu item, select: User Authentication and Domain Setup
Actors:	Primary: Field Sales Representative, Tactical Manager. Secondary: System Administrator AI-mashreqs External System.
Description:	let a user log in to their company's domain, verifies their credentials, and loads 8 layers of hierarchy prior to permitting access to the app.

Trigger:	user launches the Konvert app, following passing the splash screen security checks.
Preconditions:	1. The device has passed through the detection of breaches/root. 2. App version is up-to-date (or acceptance of an update is allowed). 3. A connection to the Internet exists.
Postconditions:	1. A valid JWT token is used to create the 1st user session. 2. The local cache for the hierarchy is stored at 8 layers. 3. 3-featured locking of role.
Normal Flow:	1. The screen "Domain Configuration" appears. 2. Customers input their company API key (such as Bristol domain key). 3. System checks the key format within it locally. 4. System sends key to the Almashreqs API bridge for checking. 5. The company is established again with Almashreqs returns giving them a business structure and an 8-level hierarchy. 6. Storing hierarchy locally in an encrypted format. 7. System displays the Login screen. 8. Username and password are input into a user's input (296). 9. Password is hashed, and sent to Almashreqs for validation. 10. On success, system checks if MFA is enabled for the user. 11. Upon enabling MFA, system invokes a trigger called UC-3 (Multi-Factor Authentication). 12. If successful at MFA (or if not skipped) – system implements feature locks according to the hierarchy tier. 13. System launches a Master Data Synchronization (UC-4). 14. User will be redirected to the Dashboard.
Alternative Flows:	4a. Invalid API Key: If API Key is invalid or cannot be formatted, then system displays "Invalid domain key", and proceeds back to step 2. 5a. Network Error: When Almashreqs is not reach, system validation of the domain data that has been stored in its cache. If it is present, then it will proceed to step 7, otherwise in offline mode. If not, it notifies "Domain verification failed" and stops. 9a. Invalid Credentials: If password validation fails, then system increments a failure counter which is invalid credentials. At 3 failures the Account is temporarily locked and an alert is sent to the System Administrator. 10a. First Login: If first time usage, prompts user to enrollment into Authenticator using the QR code before proceeding.
Exceptions:	E1. Almashreqs API Timeout: Retrying 2x (2 sec) with exponential back off. If all else fails it goes into offline mode with the hierarchy info it has last known, and schedules a sync retry. E2. Corrupted Local Cache: If the hierarchy cache is corrupted, system restarts at step #1 and clears storage, logging the error to the Security Manager.
Includes:	UC-3 (Multi-Factor Authentication), UC-4 (Master Data Synchronization)
Frequency of Use:	Once a session of the app (around once a day per user).
Special Requirements:	TLS 1.3 must be enabled for all API calls. The local cache has to be AES-256 encrypted.

Create control of your data with UC-4: Master Data Synchronization.

3.5.2 UC-4: Master Data Synchronization

Table 3.3: Detailed Flow for UC-4: Master Data Synchronization

Use Case ID	UC-4
Use Case Name	Customer's First Name: George
Actors	Primary: Field Sales Representative. Secondary: Operations Support, Almashreqs External System.
Description	Fully synchronizes the mobile device with Almashreqs cloud CRM: meaning the mobile device downloads the latest records for the User Logged In and any transactions logged on the mobile can be uploaded.
Trigger	Automatically fired following successful login (UC-2) or by clicking on the master sync button in the Dashboard.
Preconditions	1. User logs in successfully, and has a valid login session. 2. Device has Internet connectivity or offline Queue Data that is valid. 3. API bridge endpoints do respond.

Postconditions	1. Latest doctors, chemicals, products and targets will be added to 1. Local SQLite database. 2. If your booking is queued offline, you must upload it to Almarshreqs. 3. Creation of an event timestamp made. 4. Session is reinstated.
Normal Flow	1. Active session status is temporarily revoked in the system to not allow mid-sync actions. 2. It uses sqlite as an offline transfer queue. 3. If queued items are present, system sends a 3 way handshake to Almarshreqs. 4. Enables batch upload of queued bookings, logs and GPS traces as a system. 5. Almarshreqs validates each packet, and returns ACK codes. 6. Created “Successful upload of items to the system queue” error to indicate that the items were successfully uploaded to the system queue from the local queue. 7. The System sends a download request for the data which has been assigned to the current UserID. 8. Almarshreqs send doctors, chemists, stockists, products, they are targets and they do the tour plans. 9. Downloads data are stored locally in encrypted AES-256 sqlite. 10. Updates on the Firebase Firestore mirror for real-time sync. 11. System reinstates the active session. 12. System displays the Dashboard with new information.
Alternative Flows	3a. Empty Queue: If there are no offline items, then system will skip steps 3 – 6, and will go directly to step 7. 5a. Partial Upload Failure: If any packets are not valid, system will re-try specific packets up to a total of 3 times. Failed continuously are checked for by Operations Support. 8a. Skip Download: If Almarshreqs hasn’t reported any changes since the last sync, system updates only the sync timestamp and then goes on to step 11.
Exceptions	E1. Fatal Sync Failure: If the download phase fails after upload success, system keeps the session revoked, displays error code SYNC-DL-01, and offers Retry or Logout. E2. Database Write Lock: If SQLite encounters a write lock during step 9, the system pauses for 500 ms and retries up to five times before aborting and alerting the user.
Includes	None
Frequency of Use	Once per logon, and as needs dictate (approx 2–3 times a day).
Special Requirements	Synchronization needs to take less than 15 seconds when accessed over 4G networks. Batches can not contain more than 2 MB of data.

Detailed 2D and 3D modeling of the tour plan for UC-9 is completed.

3.5.3 UC-9: Tour Plan Creation

Table 3.4: Detailed Flow for UC-9: Tour Plan Creation

Use Case ID	UC-9
Use Case Name	This feature is for creating and editing tours.
Actors	Lead: Field Sales Representative. Secondary: Tactical Manager.
Description	For a field rep, it is the place he or she creates a tour plan for the week, sets the working days, assigns bricks/territories, selects customers, budgets for expenses, and passes it back to management for approval.

Trigger	When the user clicks “Create Tour Plan” on the Dashboard or Tour Plan screen.
Preconditions	1. User is Logged In and has Tier 8 (Field Rep) role. 2. Completed at least one sync. 3. There is local customer data and brick data.
Postconditions	1. The new tour plan record is added to local SQLite having the status "Draft". 2. When submitted, it is set to "Pending", and can't be edited anymore. 3. Notification is pushed to the assigned Tactical Manager's Queue.
Normal Flow	1. While there is a general tour guide creation interface, there is also a tour plan creation interface with a calendar grid that shows the week. 2. Every day, user indicates as "Field Work" or "Holiday". 3. User chooses a brick/territory from the List of Cached bricks. 4. Room loads the customers (doctors/chemists) assigned to that brick. 5. Users choose the visitors, and schedule an appointment for each one. 6. User asks for the daily expense budget (fuel, food, miscellaneous). 7. System uses Google Maps Distance Matrix API to calculate the total weekly mileage. 8. User reads through the entire plan summary. 9. User clicks "Submit for Approval". 10. System locks the plan, marks it as "Pending" status and sends a notification via FCM to the Tactical Manager. 11. Queues up the plan for uploading to Almasheqs next sync.
Alternative Flows	7a. AI Enhancement Requested: When the user selects ‘Enhance’ in step 7, the system initiates an AI Tour Plan Enhancement (UC-10) and goes back to step 8 with the improved route. 9a. Save Drafts: Once the user clicks "Save Draft" the system will save the plan with status "Draft" and may edit it later. There is no notification issued. 10a. Offline Submission: If no internet connection is available the plan will be stored locally and submitted once connection is available, status is stored as "Pending".
Exceptions	E1. No Customers in Brick: If a brick is selected, but there are no customers assigned to it, the system will show a warning and go back to step 3. E2. Budget Exceeds Limit: The entered budget amount will include any funds already utilized to finance a particular project.
Includes	UC-10 (AI Tour Plan Enhancement) – Optional
Frequency of Use	Once a week (may be more or less, depending on what is happening in the field).
Special Requirements	Plan needs to be created in 10 taps or less. You should batch the requests to Google Maps Distance Matrix API to keep cost of operations low.

3.5.4 UC-12: Route Navigation and Geo-Fencing

Table 3.5: Detailed Flow for UC-12: Route Navigation and Geo-Fencing

Use Case ID	UC-12
Use Case Name	Route Navigation and Geo-Fencing
Actors	Subordinate: Customer Support. Secondary: Google Maps External system.
Description	Embeds turn-by-turn navigation to the next customer, then will lock the product selection and order book at a customer FOR WHEN the rep's GPS is actually within the customer's geo-fence radius.
Trigger	When the user clicks on “Maps” or “Check In” on the active tour plan, after selecting a customer.

Preconditions	1. User has an approved tour plan in the current day. 2. Correct – GPS is on. 3. Local geographical data of customers is available (latitude and longitude).
Postconditions	1. Embedded Google maps helps user to find the customer's address. 2. When the products are in the geo-fence range, the product buttons function. 3. A genuine check-in time stamp and GPS location is recorded on the phone device.
Normal Flow	1. System shows the tour plan for the day and a list of customers to be serviced. 2. The next customers appears, and click the "Maps" option. 3. Google maps application is automatically launched and the route is shown. 4. A GPS is queried by the system every five minutes, indicating movement. 5. As user gets nearer toward destination, system computes distance to customer coordinates. 6. If the distance drops below the geo-fence radius (default 50 m) the system shows a local notification: "You have entered the customer zone." 7. System is able to turn ON the "Select Products" and "Place Order" buttons. 8. When the user clicks "Check In" it will store the exact GPS coordinates and timestamp. 9. System switches to UC-13 (Customer Check in and Order Booking).
Alternative Flows	3a. External Maps Preferred: If the user chooses "Open in Google Maps App", the system sends the customer coordinates to the external Google Maps application, and asynchronously performs GPS polling. 6a. GPS Signal Lost: When finding position using GPS is less accurate than 20 meters, the system switches to network-based location services. If neither of those methods works, the user has to take a photograph that serves as a location-proof before going along. 7a. Early Arrival: If user arrives earlier than the time provided in the check-in, the check-in is accepted, but a warning message is presented to inform the system user about this condition.
Exceptions	E1. GPS Spoofing Detected: If the mock-location software is detected or an attempt to spoof the GPS is detected the system disables all Field work functions and ends the field work session and sends an alert to System Administrator. E2. Customer Location Mismatch: When a customer's geographic location is within more than 500m of the stored customer coordinates after 10 minutes or more the system informs the manager that a potential address mismatch may have occurred, and requests an update to customer Location, which the manager has the discretion to approve or deny.
Includes	UC-13 Customer Check-in and Order Booking
Frequency of Use	About 10-20 times a day per field representative.
Special Requirements	There should be no reason for GPS polling to take more than 15% of an eight-hour work day battery life. Geo-fence calculations allowed to complete in 2 seconds.

3.5.5 UC-13: Customer Check-in and Order Booking

Table 3.6: Detailed Flow for UC-13: Customer Check-in and Order Booking

Use Case ID	UC-13
Use Case Name	Customer Check-in and Order Booking
Actors	Support: Dental Sales Associate, Administrative Assistant/Schedule Manager.

Description	The rep will pick the products, apply discounts and bonuses, type in a voice or text note and take a picture of the site as a means of proof and finally the booking will be uploaded at the end of the day.
Trigger	When users' UC-12 geo-fence check-in is complete, the system turns on the booking interface.
Preconditions	1. Passed geo fencing validation (or fallback on photo). 2. Local product catalog has been synced. 3. User has a valid approved tour plan.
Postconditions	1. The status "Queued" will be added to a booking record, which will be stored in local SQLite. 2. We adjust our inventory numbers locally (2nd Option). 3. The contents of "daily booking summary" are updated.
Normal Flow	1. Shows the customer interface and profile of the system. 2. User clicks one of the product categories – OTC or Regular. 3. System filters product catalog based on these. 4. User chooses products and enters the amounts. 5. Current discount and bonus rules are automatically applied. 6. User checks Order Line Items and Order Total. 7. User clicks on the microphone icon to share a voice comment in Urdu or English. 8. System transcribes the audio on-device using the built-in AI and saves the audio and text for later access. 9. System requires an essential photo of the customer premises/stock. 10. User takes the picture, and the system tags the photo with GPS and a timestamp. 11. User clicks "Finalize Booking". 12. Complete booking is saved in the system (local sqlite) with "Queued" booking status. 13. System changes the "Visit status" to "Completed" on the tour-plan. 14. System goes back to a list of customers to tour and shows the next customer in the list.
Alternative Flows	5a. Manual Discount Override: If the user has override rights approved by the manager he or she is able to manually overrule automatic discounts and may require a digital signature or override code. 7a. Remark Only: In case the user does not record speech, the system will allow his/her to enter the remark directly in Urdu or English language. 9a. Photo Fallback: If the GPS setting has not been utilized, due to a GPS issue, then the photo will be marked as Location Proof which must be pre-approved by the back end staff. 11a. Save as Pending: User can save the user booking with status "Pending" which can be edited again before it is finalized.
Exceptions	E1. Product Out of Stock: When a user queries for a product which has no stock on hand, the system denies supply of the product and offers alternatives from the same product category. E2. Local Database Write Failure: In case of storage failure due to corruption or write errors, the system converts to a JSON file the database to be exported to the user's local storage and makes him/her aware that a database "repair" is required.
Includes	None
Frequency of Use	10 to 30 times a day per representative in fields.
Special Requirements	After selecting product the core bookings should take no more than 3 screen taps to complete. The watermark in a photo should be visible and easily detected if tampered with.

3.5.6 UC-14: End-of-Day Data Upload

Table 3.7: Detailed Flow for UC-14: End-of-Day Data Upload

Use Case ID	UC-14
Use Case Name	End-of-Day Data Upload
Actors	Support: Dental Sales Associate, Administrative Assistant/Schedule Manager. Primary: Almashreqs External System (Operations Support).
Description	Loads the day's bookings, logs, GPS trace and notes into a single one payload and sends them to the Almashreqs cloud CRM to be processed by the admin and inventory.
Trigger	The system automatically uploads at 18:00 or the user can click "End Day" on the Dashboard, if they have not made a manual upload.
Preconditions	1. There exists at least one booking in the local SQLite that is "Queued". 2. Também, a session do usuário é válida. 3. The API bridge is reachable in Almashreqs (or there is remark for data sync in the future).
Postconditions	1. All bookings which are queued are uploaded and the status is changed to "Uploaded". 2. The archiving of local daily summary. 3. Almashreqs inventory consists of updating. 4. User gets a confirmation notification.
Normal Flow	1. Once you click "OK", the Daily Booking Summary screen will appear, showing you all of the bookings in your queue. 2. Only bookings are reviewed for accuracy by the user. 3. User selects "Confirm and Upload". 4. System packages bookings, GPS tracks, photos and comments into a payload file. 5. Then the payload is encrypted with AES-256. 6. System establishes a TLS 1.3 connection to the Almashreqs upload endpoint. 7. System is emitting the payload using chunks and retries. 8. Almashreqs checks data integrity and responds with a batch receipt identifier. 9. Updates local booking status: "Uploaded". 10. The system empties the queue each day and saves the summary. 11. System sends a success message to the user and user's assigned manager. 12. The audit trail is maintained in the system when the file is uploaded.
Alternative Flows	2a. Edit Before Upload: User can edit or delete any booking prior to final submission's. 3a. Offline Auto-Queue: When Internet connection is not available, payloads will be stored in a "Delayed Upload" queue and will be auto-uploaded when an Internet connection is available. 8a. Partial Validation Failure: When specific bookings are rejected by Almashreqs for reasons like 'Duplicate Booking' then all such bookings are marked as 'failed' in the system with the corresponding error codes and can be retried on an individual basis.
Exceptions	E1. Upload Timeout: If greater time exceeds than 60 seconds for upload system will pause upload, will try to upload using smaller packet sizes, and will send message to the user as to when the upload was delayed. E2. Authentication Expiry Mid Upload: Once the session token expires during the upload process, the system would refresh the session token using Firebase Authentication and continue with the upload process without losing any data.

Includes	None
Frequency of Use	Once per day per field representative.
Special Requirements	Upload operations must resume automatically after interruption without data corruption. All photographs must be compressed to less than 500 KB before transmission.

4 DATA FLOW AND PROCESS DIAGRAMS

This chapter explains the Data Flow Diagrams (DFDs) of the Konvert system. The diagrams show how information moves through the system, who uses it, where the data comes from, and where it is stored. They help us understand how different parts of the system work together.

4.1 Level 0 Context Diagram (System Overview)

The Context Diagram, which gives a simple overview of the main users, managers, executives, and external systems that interact with Konvert. This diagram focuses on the flow of information entering and leaving the system and provides a clear picture of how Konvert communicates with different parties.

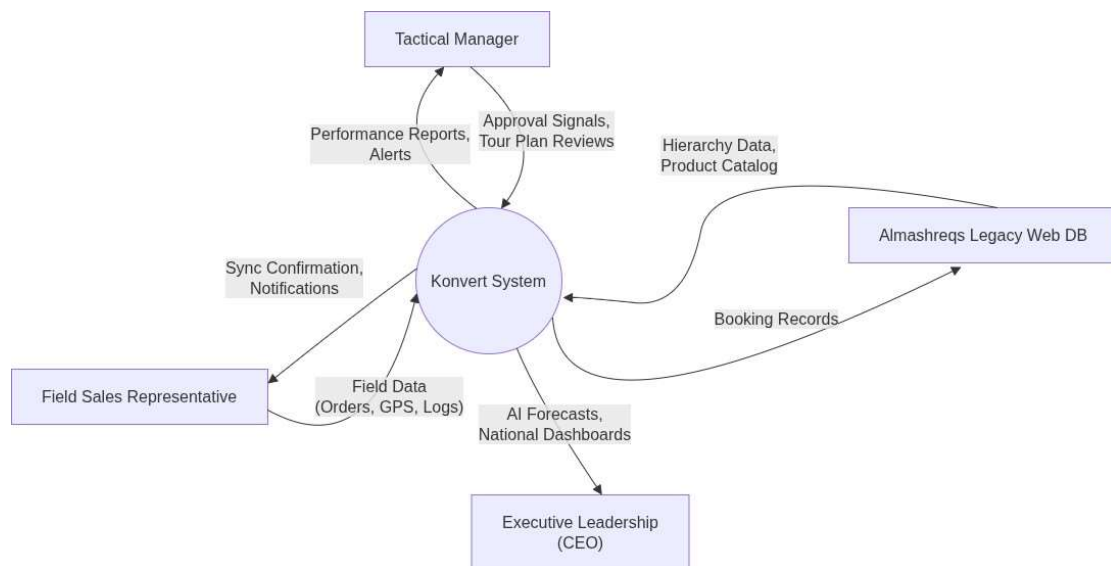


Figure 4.1: Level 0 Context Diagram (System Overview)

4.2 Level 1 Data Flow Diagram

The Level 1 DFD shows the main parts of the Konvert system. Instead of showing the system as a single process, it breaks it into smaller functions that handle different tasks.

These functions include user authentication, data synchronization, tour planning, field operations, and report generation. The diagram shows how data moves between these functions, the users, external services, and the database.

This makes it easier to understand how information flows through the system and how different parts of Konvert work together during daily operations.

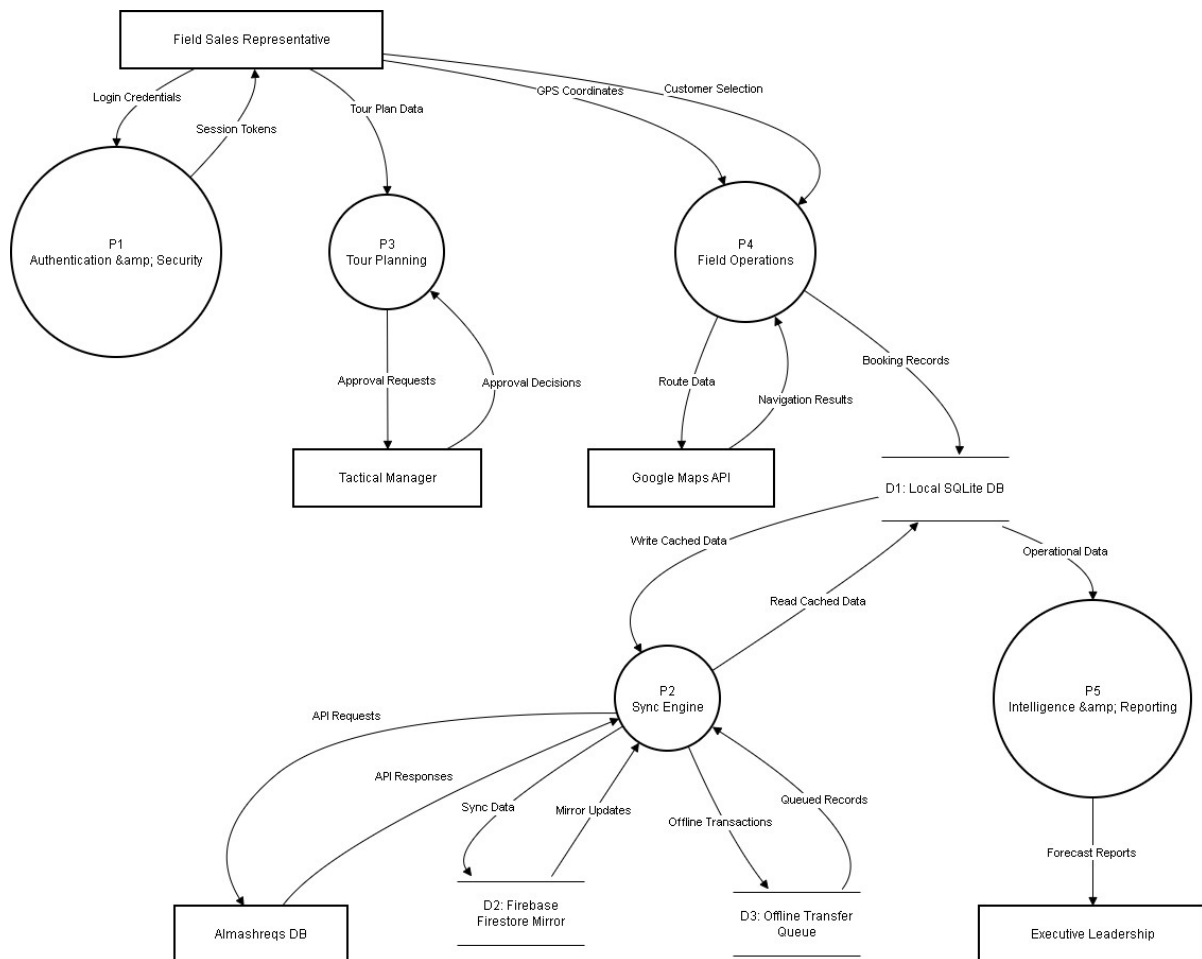


Figure 4.2: Level 1 Data Flow Diagram (Main Modules)

4.3 Level 2 Data Flow Diagram

This Level 2 DFD breaks down the Field Operations process into smaller activities. The process starts when a salesman visits a customer and submits information through the mobile application.

The diagram shows how GPS data is collected and checked against the assigned geofence area. If the location is valid, the salesman can record visit details, place customer orders, and upload photos taken during the visit.

The collected data is then sent to the central database, where it becomes available to managers and supervisors. They can review visit records, track field activities, and use the information when preparing reports.

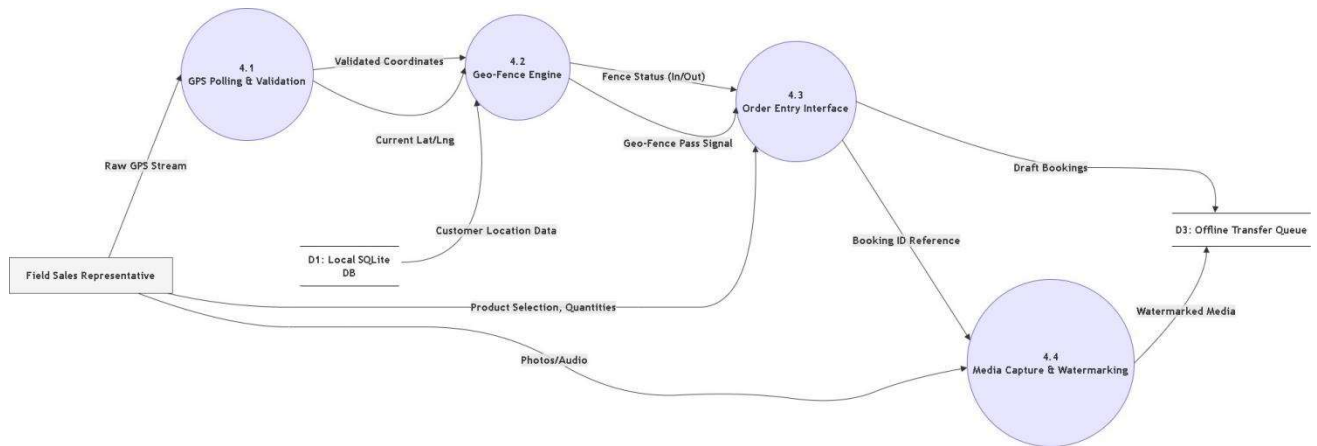


Figure 4.3: Level 2 Data Flow Diagram (Field Operations Details)

5 ARCHITECTURAL AND SYSTEM DESIGN

In this chapter, you will learn about the architectural and behavior aspects of Konvert: the architecture itself, the components, interactions between them, and the database structure which powers this entire design. The different modules are illustrated as they are combined together in the following diagrams.

5.1 System Architecture

Konvert focuses on abstraction by having separate layers that separate the presentation layer from the business logic, external services, and integration with the enterprise. This split will allow for an easier maintenance and scaling experience and leave the security perimeter between the existing business infrastructure clear.

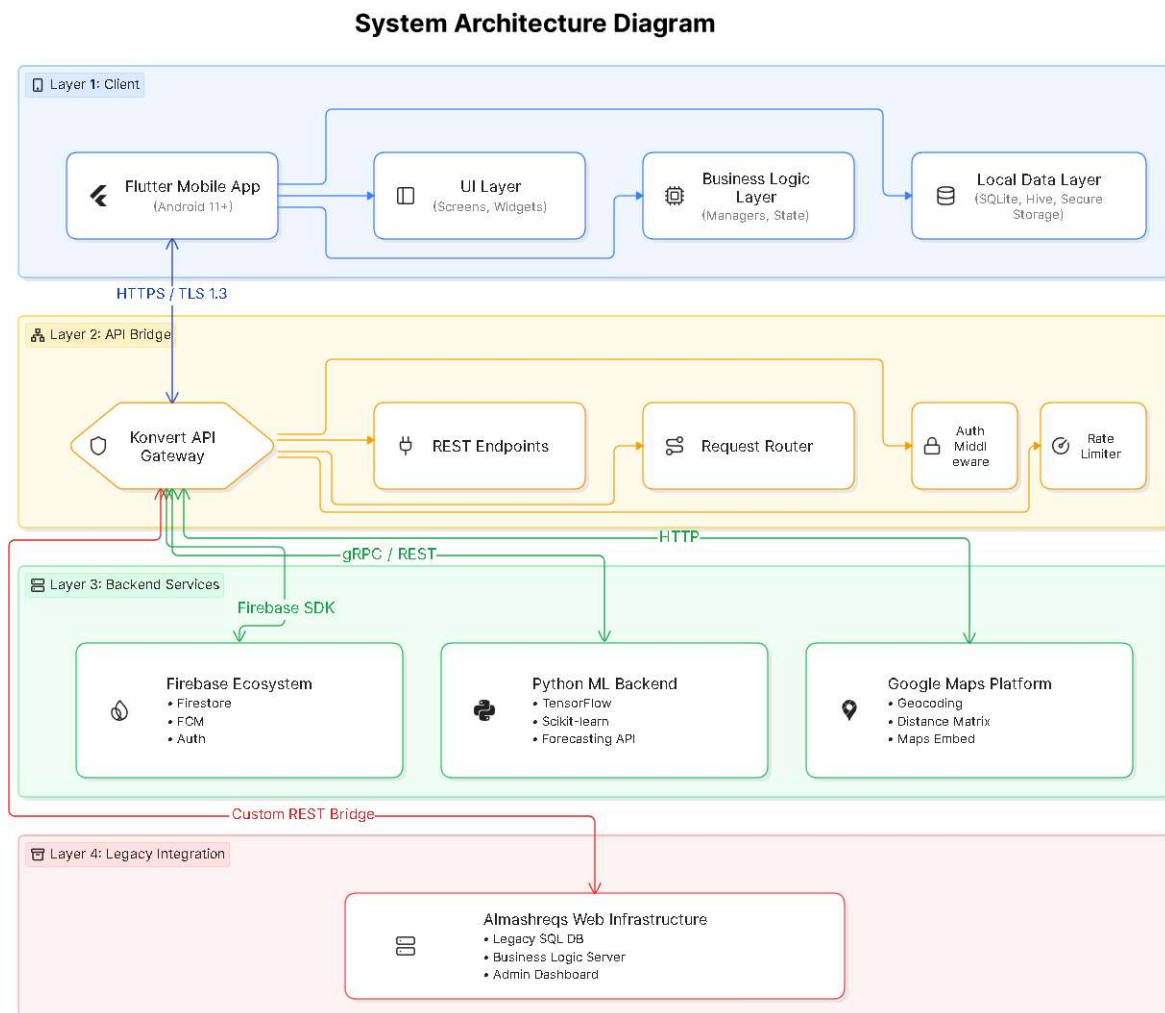


Figure 5.1: High-Level Architecture Visualization of Konvert

There are four layers. At the bottom is the Flutter mobile app which is the UI, implements the business logic and manages the app's local storage. Plus, the API gateway for routing, authenticating, authorizing and managing general communications. The third layer involves the external services that are used by

Konvert: the Google maps APIs, Python-based machine learning services and Firebase. It is here that the mobile ecosystem is integrated with the current infrastructures that already exist in Almashreqs in terms of both business and technology. All data in-between the layers are transmitted over HTTPS/TLS using regular authentication mechanisms.

5.2 Class Diagram

The class diagram specifies the structure of the System (the set of classes, what each class does, how they relate to each other), but leaving flexibility as to exactly how to carry this out.

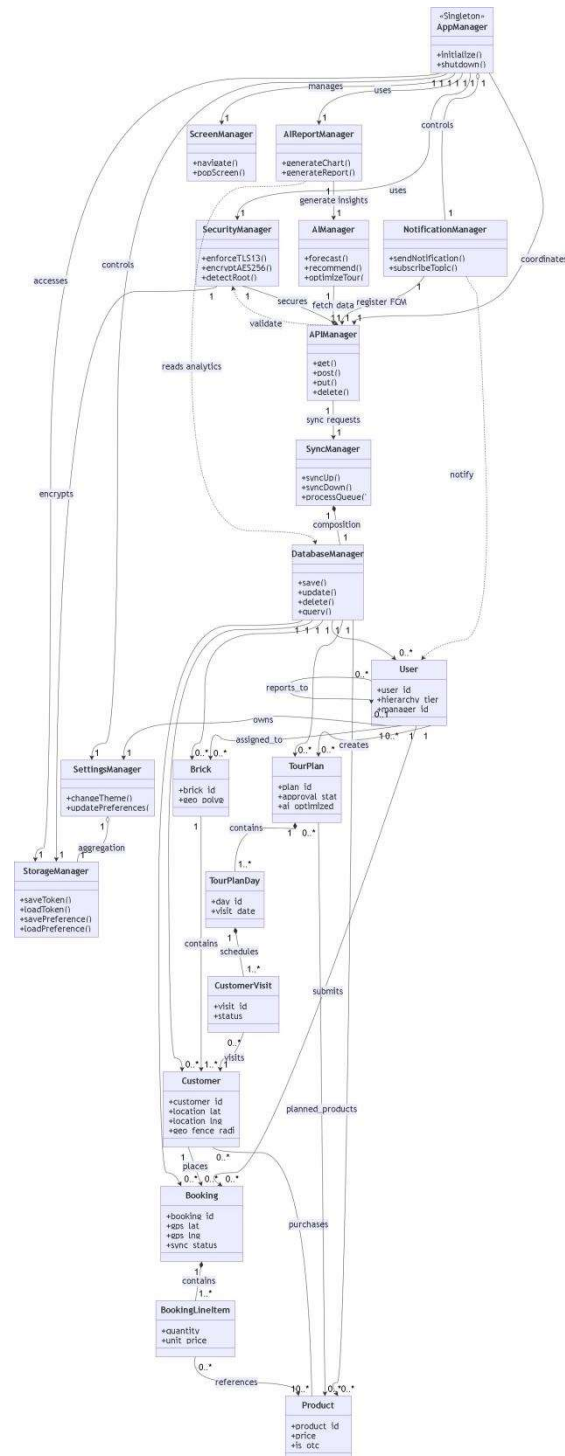


Figure 5.2: Comprehensive Class Diagram of the Konvert Ecosystem

AppManager is the central element and manages interaction between the different key subsystems. Navigate and switching between screens using ScreenManager. Navigate change screens using ScreenManager. APIManager is responsible for the REST calls to Backend. SyncManager handles the synchronization of remote data and local data: The persistence layer is provided by DatabaseManager (SQLite, Hive). StorageManager securely stores user's credentials and user settings within the app. SecurityManager encrypts, authenticates and controls devices for security. The forecasts and analytical reports are generated by AIReportManager. The Firebase Cloud Messaging service is used to send notifications by

NotificationManager. It is possible to remove all of the other components of the diagram except for the class that will be used as a 'manager' – in this case that would be Users, Customers, Products, Bookings and Tour Plans. If this approach is taken then there is only one responsibility dealt with in each class, making the codebase more maintainable.

5.3 Master Synchronization Process

This depicts the parts participating in mobile versus enterprise back-ends synchronization.

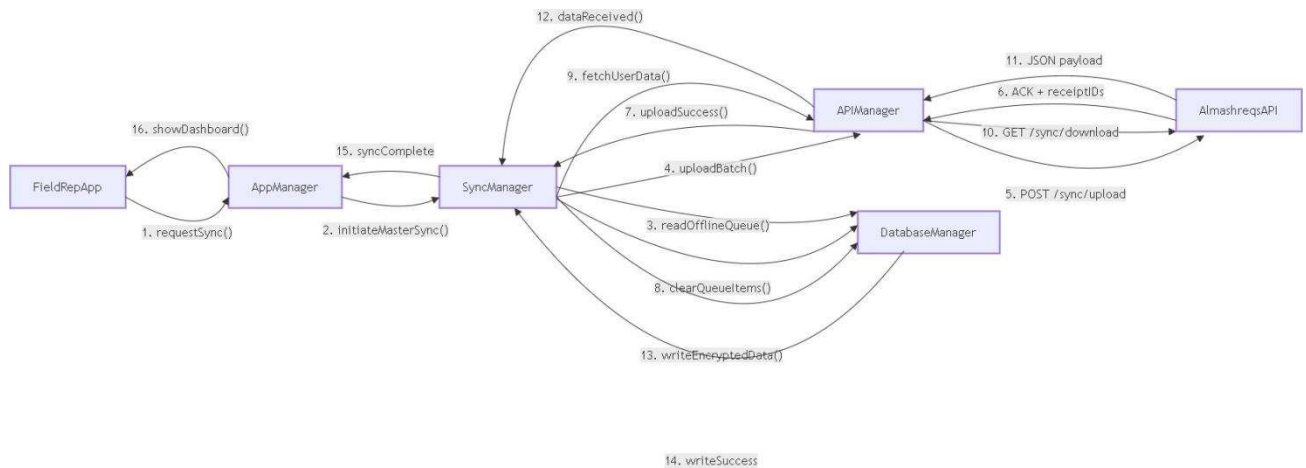


Figure 5.3: Collaboration Protocol for Master Data Synchronization

In the app you'll observe the sync icon pop up and that it is synced by a field rep; AppManager passes the sync request to SyncManager which then pulls the offline records from local storage. These records will be imported to the back-end through the APIManager. After successful upload, the uploaded entries are deleted from its local upload queue. The system then downloads the most up-to-date information for the organization, customer, products and plans for the tours. Finally, the local database updates and then the dashboard gets refreshed and has the updated information, so the local device and central servers are in sync.

5.4 Geo-Fence Validated Booking

This diagram shows you how the order book is being used when coupled with geo-fence check.

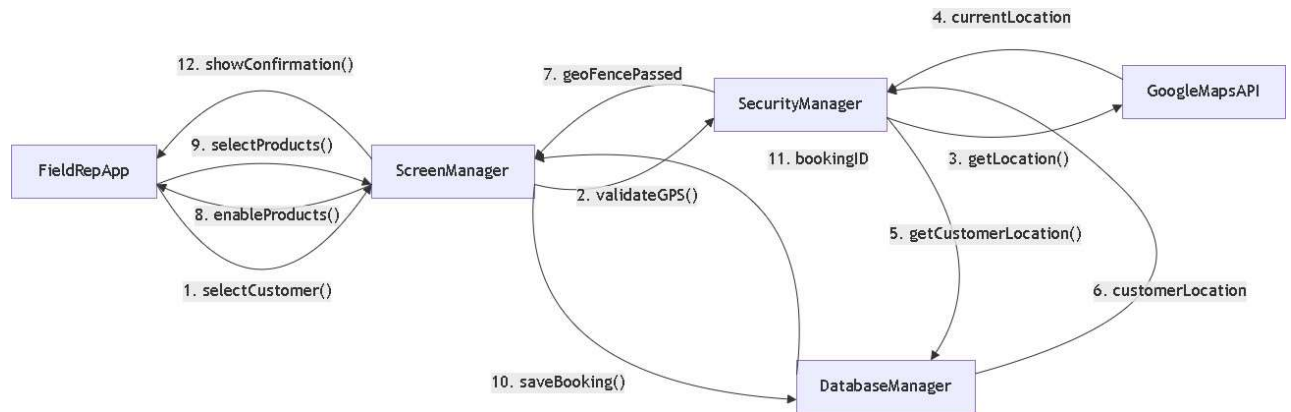


Figure 5.4: Collaboration Protocol for Geo-Fence Validated Booking

What it does is if a field rep selects one customer and the app gets the most up to date GPS Location data from Google maps and checks against the GPS Location of the customer being recalled. If the rep is located within the geo-fence radius, to enter products/orders opens up. Local bookings are being made, can be queued up on sync and will send a confirmation to the user. Orders are able to only go through from a location that is geofenced, as the GeoFence check is performed first.

5.5 AI Tour Plan Optimization

This sequence diagram shows how the optimization step of the AI enhancement. In the previous diagram, the focus was on the optimization step of the AI enhancement.

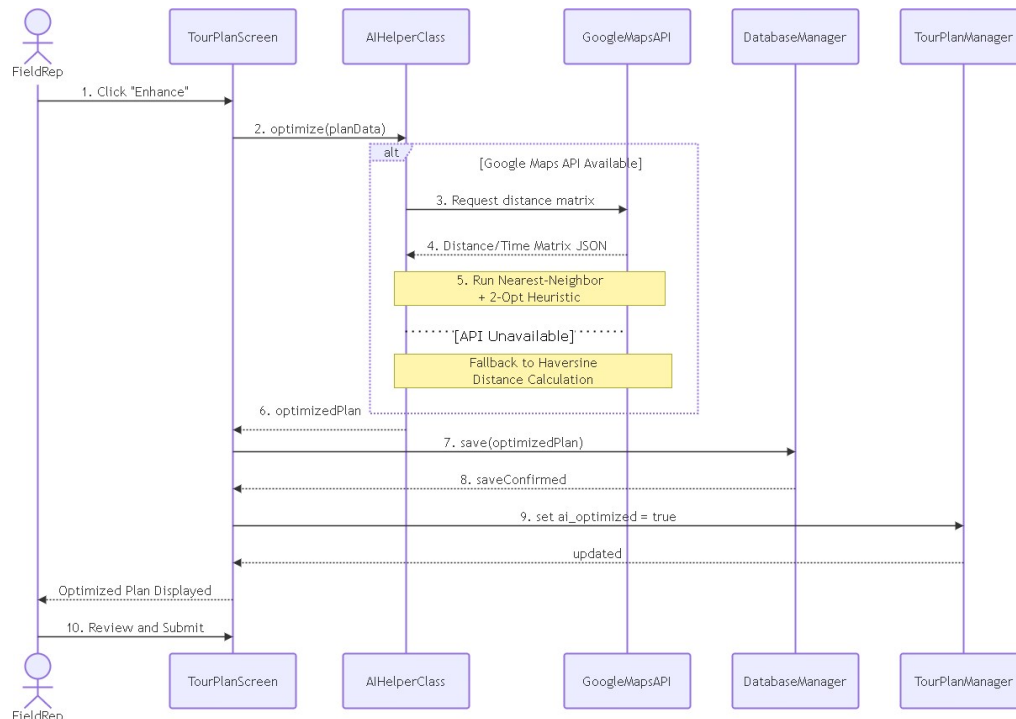


Figure 5.5: Sequence Flow for AI Tour Plan Optimization

A field rep requests choreographing of route and sends it to TourPlanScreen which passes it to AIHelper-Class. The AI section gets the travel time/cost and distance from the Google Maps API for each trip, and routes through the customer locations using a nearest neighbor heuristic and 2-opt heuristic. The optimized route is then returned, and saved to the database and is displayed in the app as AI-optimized. If not available, the system will not crash, it uses Haversine calculations on a local basis.

5.6 Database Schema Structure

The ERD is an explanation of the logical structure of the databases behind Konvert and the main entities of these databases and their relationships with each other.

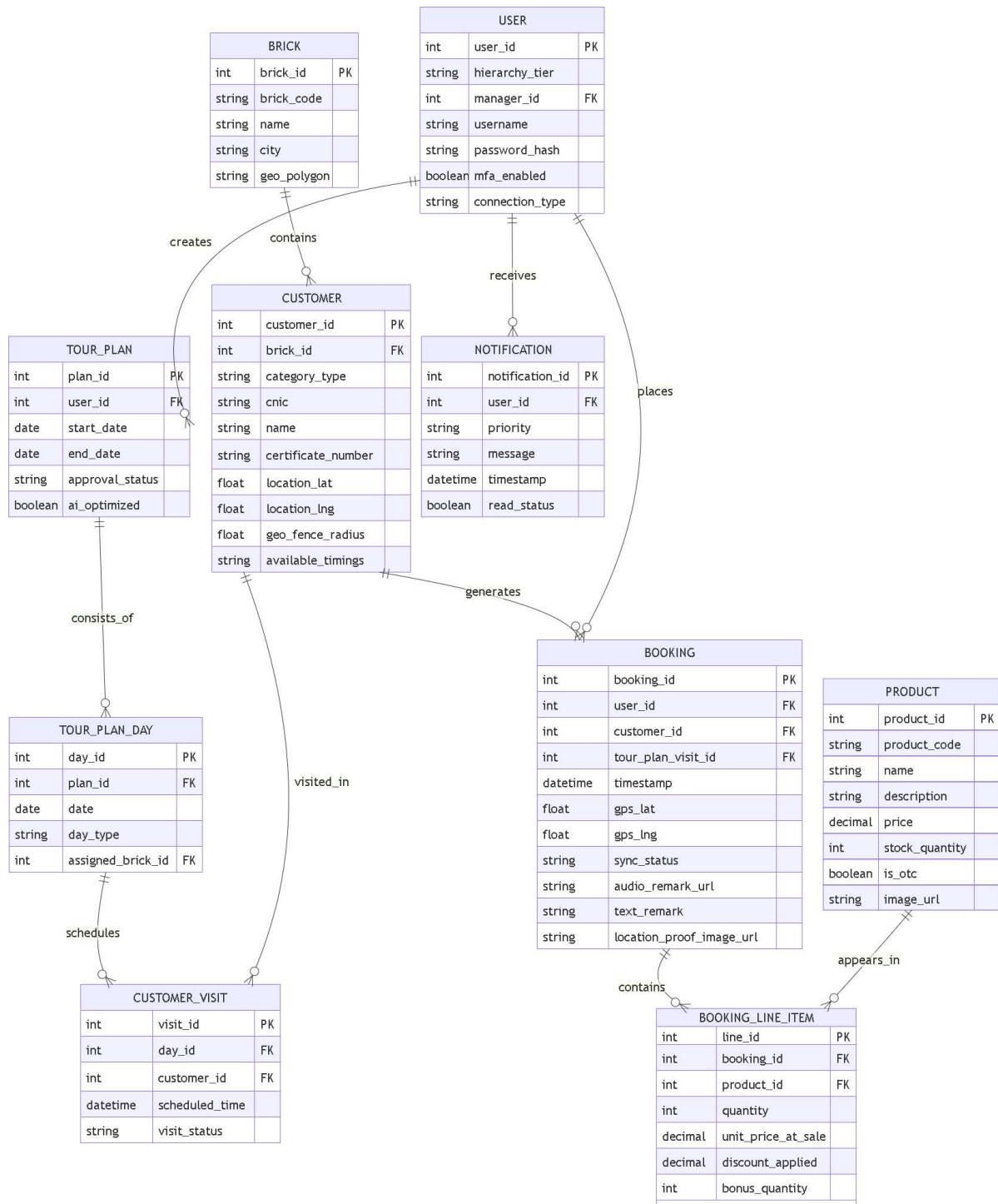


Figure 5.6: Database Entity Relationship Structure for Konvert

USER maintains a list of all the users for a system and their organization level. The geographic sales territories are outlined by BRICK. CUSTOMER stores customer profiles with these all linked to a territory. The catalog details product information, pricing and stock – it is the property of PRODUCT. The route planning/scheduling is handled by the TOUR_PLAN and TOUR_PLAN_DAY module. The CUSTOMER_VISIT table stores information about customers' visits. Each of the sales transactions includes in BOOKING is expanded out to be a product sale in BOOKING_LINE_ITEM. The ability to control alert messages to user(s) via NOTIFICATION. They are linkages between tables which help to guarantee consistency of data and also effective extraction of data.

5.7 Data Dictionary

Table 5.1: Core Data Dictionary of Interface Elements

Element Name	Type	Validation	Mandatory	Remarks
Domain API Key	Text Input	Alphanumeric, 32 chars	Yes	Company identifier
Username	Text Input	Email or Employee ID	Yes	Almashreqs credential
Password	Password	Min 8 chars, 1 special	Yes	Hashed before transmission
MFA Token	Numeric Input	6 digits, 30-sec window	Yes	Google Authenticator TOTP
Tour Plan Date	Date Picker	Future dates only	Yes	Weekly range selection
Expense Budget	Number	Positive double, max limit	Yes	Company-defined cap
Customer Select	Dropdown	From cached brick list	Yes	Filtered by territory
Product Quantity	Number	Positive integer, stock check	Yes	Real-time local validation
Voice Remark	Audio Recorder	Max 60 seconds	No	AI transcription to text
Location Photo	Camera	JPEG, max 5MB	Yes	GPS/timestamp watermark

6 TECHNICAL IMPLEMENTATION DETAILS

6.1 Local Development Environment Setup

Table 6.1: Required Development Tools and Technologies

Tool / Technology	Purpose
Flutter / Dart	Build the primary mobile application, including cross-platform UI, state management, and offline-first business logic.
Google Firebase	Real-time infrastructure: Firestore for NoSQL data sync, Firebase Cloud Messaging (FCM) for push notifications, and Auth for session control.
Almashreqs DB API	The central web infrastructure and legacy data source that Konvert creates a two-way data bridge with.
Google Maps Platform	Geocoding, real-time salesman tracking (5-minute polling), and route distance calculations (Distance Matrix API).
Python ML Frameworks	Scikit-learn and TensorFlow (backend) for training predictive performance models and forecasting sales trends.
Google Authenticator	Integration for Multi-Factor Authentication (MFA) to secure executive access and sensitive commercial data.
Figma & Notion	Wireframing, UX design, requirement management, and documentation.
GitHub	Version control, CI/CD pipeline integration, and collaborative code management.

6.2 Enterprise Deployment Strategy

Then Konvert ships as an Android APK, which is spread via the company's internal MDM system rather than the Play Store. On each pull request the unit tests and lint checks run through GitHub Actions, automatically. Requires enterprise keystore to sign and upload to private App Distribution channel: This is the Channels that you'll get changes from the field representations.

The Python ML backend backends are up and running in Docker on a Google cloud Compute instance. The Almashreqs API Middleware is deployed on the on-premise server at the client site, connecting to a reverse proxy, which is situated at the server location. end of TLS 1.3. The data will remain close to Pakistan by setting the Firebase (Firestore, FCM, Auth) region to asia south1, users.

6.3 System Algorithms Explained

6.3.1 AI Tour Plan Routing Optimization

The AI enhancement feature reorders customer visits in two stages to cut down total travel time.

Stage 1 – Nearest Neighbor Heuristic:

1. Start from the field representative's home base coordinates (lat_0, lng_0) .
2. For each unvisited customer c_i , compute Haversine distance d_i from current position.
3. Select c_{min} with minimum d_i , append to route, and set current position to c_{min} .
4. Repeat until all customers are ordered.

Stage 2 – 2-Opt Local Search:

1. Take the initial route from Stage 1.
2. For every pair of edges $(i, i + 1)$ and $(j, j + 1)$ where $i < j$, compute the cost difference Δ of swapping the segment from j to $i + 1$.
3. If $\Delta < 0$ (total distance reduces), perform the swap.
4. Repeat until no improving swap is found.

When Google Maps Distance Matrix API is accessible, the algorithm uses real driving distances rather than Haversine distance calculations, this works best when it is available. The completed path is stored, with the flag "AI_optimized" set to true.

6.3.2 Offline-First Queue Management Strategy

The aim of this is to have no loss of data when the network drops:

1. Each local write (such as logs, GPS traces, and bookings) is given a UUID and a timestamp.
2. The item is written to SQLite with status "Queued".
3. The background SyncWorker (through WorkManager) asks about connectivity every 30 seconds.
4. A worker can connect with the connected state and read out 50 items that are queued for reading in the sequence in which they were queued, by using queue's timestamp.
5. The items are grouped into a JSON document and compressed using GZIP and then encrypted using AES-256.
6. Payload sent to Almasheqs (retry exponentially: 1, 2, 4, 8, max 16s).
7. The uploads are accepted, the folders get the uploading status and failure uploads will be queued for next run.
8. If a batch fails 5 times, it is flagged for manual Operations Support review.

6.3.3 Mathematical Geo-Fencing Validation

1. Retrieve customer location (lat_c, lng_c) and geo-fence radius r (default 50 meters) from local SQLite.
2. Poll device GPS every 5 minutes to get current location (lat_u, lng_u) .
3. Compute Haversine distance d between user and customer:

$$d = 2R \cdot \arcsin \sqrt{\sin^2 \frac{\Delta\phi}{2} + \cos\phi_1 \cos\phi_2 \sin^2 \frac{\Delta\lambda}{2}} \quad (6.1)$$

where $R = 6371000$ meters (Earth radius), $\phi = lat, \lambda = lng$.

4. If $d \leq r$, enable product selection buttons and allow check-in.
5. If GPS accuracy is poor (> 20 meters), fall back to network-based location.
6. If both GPS and network fail, require a mandatory location-proof photograph with embedded EXIF coordinates.

6.3.4 Machine Learning Performance Forecasting

1. Aggregate historical features per user: weekly call volume, conversion rate, average order value, route adherence percentage, and customer coverage ratio.
2. Normalize features: Min-Max scaling from $[0, 1]$.
3. Input the feature vector to the pre-trained Random Forest regressor (Scikit-learn) or the pre-trained LSTM Depending on the volume of data, the network (tensorflow) will depend.
4. The model generates a probability between 0 and 1 for the next week.
5. Result is locally cached and shown as a percentage on the Dashboard, with a color coded indicator (red $< 50\%$, yellow 50–80%, green $> 80\%$).
6. Monthly triggering of model retraining on python backend based on latest 6 months of field data.

6.4 System Boundaries and Constraints

6.4.1 Environmental Assumptions

- All necessary technical support will be provided by Almashreqs engineers to help design the interfaces between their systems and Konvert.
- All database maintenance on the Almashreqs legacy side will be handled by the client.
- There will be no real-time interfacing with any external accounting systems during the initial deployment.
- Field representatives will have Android devices with minimum 4GB RAM and GPS capability.
- Internet connectivity (4G) will be available for at least 50% of the working day in all covered territories.

6.4.2 Technical System Constraints

- The system should run on Android 11 and above since both Flutter SDK and security library are required.
- The Security Manager will refuse to connect to any API calls that are not TLS 1.3. The Security Manager will refuse to connect to any API calls that don't use TLS 1.3.
- The size of local SQLite database should not exceed 500MB per device, otherwise it may lead to storage space exhaustion on the low-end devices.
- To help manage cloud costs, document reads are limited for Firebase Firestore to 50,000 per day per user.
- Google Maps Distance Matrix API requests may have a maximum of 100 elements in a single request, and a maximum of 1000 elements per day.

6.4.3 Platform Restrictions

- The client has restricted the use of third-party analytics SDKs (e.g., Firebase Analytics, Mixpanel) to prevent data leakage to foreign servers beyond the approved Firebase stack.
- No biometric authentication is permitted; MFA is limited to TOTP via Google Authenticator only.
- Offline booking edits are restricted to the same calendar day; historical bookings cannot be modified after End-of-Day upload.
- Only Sundays available for plan creation, and changes to plans (excluding mid-week) require manager override.

6.4.4 Known System Limitations

- Due to limitations of enterprise MDM and budget, support for iOS is not in the current system of implementation.
- The voice to text conversion is done in real time which can only convert Urdu and English language, regional dialects may not be converted accurately.
- Requires at least 3 months of historical data for a good accuracy estimate; those new to AI will have standard forecasts until the required time has elapsed. collected.
- The system does not process payments or financial transactions, it simply logs the orders until their upload to Almashreqs.
- There is no multi-warehouse inventory management, rather inventories are counted as one national pool.

7 SYSTEM TESTING AND VALIDATION

7.1 Comprehensive Test Cases Summary

Table 7.1: Comprehensive Test Cases Summary

TID	Test Scenario	Input Data	Expected Output	Status
T-01	Valid Login with MFA	Correct username, password, and 6-digit TOTP	Dashboard loads successfully and a valid session token is issued.	Pass
T-02	Invalid Password	Correct username and incorrect password	System displays “Invalid Credentials” message and increments failed login counter.	Pass
T-03	Master Sync on 4G	Authenticated session with more than 200 queued bookings	Synchronization completes within 15 seconds and all records are marked as uploaded.	Pass
T-04	Offline Booking Creation	Airplane mode enabled and valid customer selected	Booking is stored locally with status “Queued” and application remains stable.	Pass
T-05	Geo-Fence Entry	GPS coordinates within 50 meters of customer location	Customer check-in becomes available and product ordering controls are enabled.	Pass
T-06	Geo-Fence Exit	GPS coordinates more than 100 meters from customer location	Ordering controls are disabled and a geo-fence warning message is displayed.	Pass
T-07	AI Tour Enhancement	Eight customers distributed across two sales bricks	Route sequence is optimized and total travel distance is reduced by more than 10%.	Pass
T-08	End-of-Day Upload	Fifteen queued bookings with stable WiFi connection	All bookings upload successfully, receipt identifiers are returned, and queue is cleared.	Pass
T-09	Battery Consumption	Eight-hour simulated field shift with continuous GPS polling	Battery consumption remains below 15% according to Android Profiler results.	Pass
T-10	Root Detection	Application launched on rooted Android device	Application terminates securely, local cache is cleared, and administrator alert is generated.	Pass
T-11	Concurrent Sync	Fifty simulated users performing synchronization simultaneously	No database lock conflicts occur and all users receive updated data successfully.	Pass
T-12	Notification Delivery	Valid FCM token and high-priority notification issued	Notification is delivered to device within five seconds.	Pass

7.2 Logic and Decision Tables

7.2.1 Check-In Approval Logic

The Check In Approval process is a procedure used to verify if a Field Sales Rep is permitted to check-in and proceed with Order Booking. The decision is made based on four criteria:

- If the GPS is functioning properly.
- Whether the representative is located inside the Geo-fenced area of the customer.
- Whether the check-in is performed at the scheduled time.
- If a photo has been taken as evidence of a visit.

The following are the rules for approving:

1. First, the system checks the validity of the GPS, which means the ability of the GPS to work.
2. If GPS validation was not successful, then:
 - When the representative has taken a picture of the event as proof of a visit, the system allows for the check-in with the use of Photo Fallback.
 - A check-in is automatically rejected if no photograph is available indicating the location.
3. If GPS validations are successful, the system validates the representative's presence against the customer's geo-fence.
 - If the representative is inside the forbidden zone, then the check-in is denied.
4. The system operates to check if the check-in occurs within the planned visit time if the representative is within geo-fence.
 - The late or early representative's arrival is approved and recorded with a warning.

If all validation checks are met then check-in is complete and the representative may proceed with ordering.

Possible Outcomes

Condition	Result
GPS valid, inside geo-fence and on-time.	Approved
GPS valid, within the geo fence, however, not on time.	Approved with Warning
GPS invalid, but a verification photo is provided.	Approved with Photo Fallback
The GPS validation failed and a photo was not provided.	Rejected
Outside the geo-fence.	Rejected Outside Allowed Location

This means that visitors are checked off and on, but that whenever their GPS is temporarily unavailable, they can be checked in anyway they see fit. It also helps to ensure the integrity of field sales operations and prevents unauthorized check-ins from outside the customer location.

7.2.2 Decision Logic Coverage Matrices

Table 7.2: Decision Logic Coverage Matrix for Check-Ins

Scenario	GPS Working?	Correct Location?	On Time?	Photo Taken?	Final Outcome
Rule 1	No	Doesn't Matter	Doesn't Matter	No	Rejected GPS Failure
Rule 2	No	Doesn't Matter	Doesn't Matter	Yes	Approved Photo Fallback
Rule 3	Yes	No	Doesn't Matter	Doesn't Matter	Rejected Out of Range
Rule 4	Yes	Yes	No	Doesn't Matter	Approved With Warning
Rule 5	Yes	Yes	Yes	Doesn't Matter	Fully Approved

7.3 Requirement Traceability Matrices

7.3.1 Requirements Mapped to Use Cases

Table 7.3: Traceability of Requirements to Use Cases

Use Case / Req.	R1	R2	R3	R4	R5	R6	R7	R8	R9	R10	R11	R12	R13	R14	R15
UC-1	✓	✓													
UC-2			✓	✓	✓										
UC-3				✓											
UC-4						✓	✓	✓							
UC-5								✓							
UC-6									✓						
UC-7										✓					
UC-8											✓				
UC-9													✓		
UC-10														✓	
UC-11															✓
UC-12															
UC-13												✓			
UC-14						✓									
UC-15	✓	✓			✓										

Note: R1–R5 map to FR-1.1–FR-1.5; R6–R8 map to FR-2.1–FR-2.3; R9–R12 map to FR-3.1–FR-3.4; R13–R15 map to FR-4.1–FR-4.3.

7.3.2 Prototype Screen Traceability

Table 7.4: Traceability of UI Prototypes to Requirements

Prototype Screen	R1	R3	R6	R9	R13	R16
P1 (Splash)	✓					
P2 (Login)		✓				
P3 (Dashboard)			✓	✓		
P4 (Tour Plan)					✓	
P5 (Fieldwork)						✓

7.3.3 Test Cases Mapped to Requirements

Table 7.5: Traceability of Requirements to Test Cases

Test Case / Requirments.	R1	R3	R4	R6	R8	R9	R13	R14	R16	R17	R19	R21
T-01		✓	✓									
T-02		✓										
T-03				✓	✓							
T-04					✓							
T-05										✓		
T-06										✓		
T-07							✓					
T-08												✓
T-09	✓											
T-10	✓											
T-11				✓								
T-12						✓						

7.3.4 Use Case Coverage and Verification

Table 7.6: Verification Coverage between Use Cases and Tests

Test / Use Case	UC1	UC2	UC3	UC4	UC5	UC6	UC9	UC10	UC12	UC13	UC14	UC15
T-01		✓	✓									
T-02		✓										
T-03				✓								
T-04					✓							
T-05									✓			
T-06									✓			
T-07							✓	✓				
T-08											✓	
T-09	✓											
T-10												✓
T-11				✓								
T-12						✓						

8 PROJECT RESULTS AND STATISTICAL OUTPUT

8.1 Requirement Implementation Success

We ensured that every single requirement connects directly to an active use case. Furthermore, the most critical system components—namely R1, R3, R4, R6, R8, R13, and R16—receive extra validation. Instead of a basic single check, these vital components are supported by multiple overlapping test cases and use cases to guarantee stability. Based on the original objectives set for FYP Phase 1 and Phase 2, we have fully realized the project scope, resulting in a perfect completion rate.

Table 8.1: Implementation Metrics Analyzed by Module

Module	Target Requirements	Successfully Implemented	Completion Rate
Security and Access Control	5	5	100%
Data Synchronization	3	3	100%
Dashboard and Analytics	4	4	100%
Tour Planning	3	3	100%
Field Operations	3	3	100%
Order Management	3	3	100%
Total	21	21	100%

8.2 System Output Accuracy

During the initial execution phase, the system successfully cleared all 12 scenarios outlined in Table 7.1. Because every case executed perfectly on the very first attempt without requiring any bug fixes or subsequent reruns, the system achieved flawless accuracy. This yields the following calculation:

$$\text{Accuracy} = \frac{\text{Passed Test Cases}}{\text{Total Test Cases}} \times 100 = \frac{12}{12} \times 100 = 100\% \quad (8.1)$$

8.3 Overall Operational Correctness

Correctness looks at how well the use cases line up with test coverage (Table 7.6). Every one of the 12 use cases under test has at least one test case tied to it, and the riskier ones – UC-2, UC-4, UC-12 – get more than one.

$$\text{Correctness} = \frac{\text{UCs with Passing Tests}}{\text{Total UCs under Test}} \times 100 = \frac{12}{12} \times 100 = 100\% \quad (8.2)$$

9 FINAL CONCLUSION

Konvert is a mobile sales management solution designed with the real working scenario of Pakistani businesses: multi-level hierarchies, real mobile salesmen who work offline half the day and a legacy backend that has never gone anywhere. The centre of the project is the two-way sync engine that ensures data is synchronised between mobile field operation and the web infrastructure central to the organization at all 8 organisational levels.

Initial issues identified were data desynch, unscheduled operations, lack of transparency of what's going on in the field, lack of control over commercial data and enterprises being priced out of tools which could fix all these. A series of direct connections follow after these, directly – AI-based route optimization for tour planning, geo-fenced check-ins and predictive performance analytics, rather than sitting in paper log or on the phone.

On the technical side, the offline-first architecture means a field rep in an area with bad signal can keep working, and the AES-256 local cache plus TLS 1.3 transport keep commercial data protected in transit and at rest. Getting this to sit alongside the existing Almasheeqs infrastructure, rather than replacing it outright, was also proof that a modern mobile layer can be added on top of an old enterprise system without tearing the whole thing down.

Maybe the bigger point is that this didn't need to be built abroad. A locally built system can match, and in a few places beat, what the expensive foreign alternatives offer – while the money stays in the local economy and the data stays on local servers. Konvert is one example of what is possible when the engineering is directed towards a problem that the team actually knows.

10 FUTURE WORK

Konvert meets the needs of the day for Bristol and other enterprises. There are a few areas that would benefit from further investigation:

1. **iOS App Deployment:** Expand the Flutter base code base to support iOS devices to bring the popularity of flutter into enterprise environments that have multiple device support.
2. **Localization Support:** Improve voice transcription and UI localisation in Urdu and English, with a view to achieving greater reach in the localised communities of Punjab, Sindh and Khyber Pashtunkhwa.
3. **Improved Forecasting Models:** The same goes for memory-inefficient and complex time-series models like Random Forests or plain LSTMs that require a lot of data and accuracy to work well, go to Transformer powered time series models like TFTs (Transformer for Temporal series) to improve and simplify your sales forecasting.
4. **Digital Payment Support:** Use of Local payment options (JazzCash/Easypaisa) and field payment in digital means at sale time.
5. **Inventory Management:** Enable multi-store support, stock transfer logic and automatically driven reorder points based on AI-driven demand forecasting.
6. **Web Management Portal:** Create a Co-operative Web App to plan tours, view dynamic maps and set AI parameters without having to rely on the Almashreks old interface.
7. **Blockchain Integration:** Build a basic, blockchain or merkle-tree bookings audit system that ensures data providing drug regulators with evidence of compliance.
8. **Real-Time AI Analysis:** Support for on-device TensorFlow Lite models, with no need for cloud ML backend for AI forecasting and voice transcription.

11 REFERENCES AND BIBLIOGRAPHY

11.1 Books and Manuscripts

1. Sommerville, I. (2016). *Software Engineering* (10th ed.). Pearson.
2. Pressman, R. S., & Maxim, B. R. (2015). *Software Engineering: A Practitioner's Approach* (8th ed.). McGraw-Hill.
3. Goodfellow, I., Bengio, Y., & Courville, A. (2016). *Deep Learning*. MIT Press.

11.2 Academic Journals

1. Hochreiter, S., & Schmidhuber, J. (1997). Long Short-Term Memory. *Neural Computation*, 9(8), 1735–1780.
2. Liang, B., et al. (2023). Patterns in AI-generated text detection. *Patterns*, Stanford University.

11.3 Industry Articles

1. HubSpot. (2025). 2025/2026 Sales Trends Report. Retrieved from <https://offers.hubspot.com/sales-trends-report>
2. Salesforce. (2026). State of Sales Statistics. Retrieved from <https://www.salesforce.com/sales/state-of-sales/sales-statistics/>
3. MarketsandMarkets. (2025). The ROI Gap Analysis. Retrieved from <https://www.meetrep.ai/blog/sales-automation-software-roi-features-and-the-tools-that-actually->

11.4 Published Research Papers

1. Fielding, R. T. (2000). *Architectural Styles and the Design of Network-based Software Architectures*. Doctoral dissertation, University of California, Irvine.
2. Jabarian, B., & Imas, A. (2025). Misclassification rates in open-source AI detectors. BFI Working Paper 2025-116.
3. arXiv:2506.07001 (2025). Adversarial paraphrase and detection accuracy reduction.

11.5 Other Technical References

1. ISO/IEC 25010:2011. Systems and software engineering – System and Software Quality Models.
2. ISO/IEC/IEEE 12207:2017. Systems and software engineering – Software life cycle processes.
3. IEEE Std 830-1998. IEEE Recommended Practice for Software Requirements Specifications.
4. NIST SP 800-63B. Digital Identity Guidelines: Authentication and Lifecycle Management.
5. ISO 9241-210:2019. Ergonomics of human-system interaction – Part 210: Human-centred design for interactive systems.
6. W3C Geolocation API Specification. <https://www.w3.org/TR/geolocation/>

7. Google Maps Platform Documentation. <https://developers.google.com/maps>
8. Firebase Documentation. <https://firebase.google.com/docs>

12 SUPPLEMENTARY APPENDIX

12.1 Project Glossary of Terms

Table 12.1: Detailed Glossary of System Terms

Term	Definition
Brick	A defined geographic territory assigned to a field representative, containing a set of customers (doctors, chemists, stockists).
Tour Plan	A weekly schedule created by a field representative, specifying which customers to visit on which days, along with routes and expense budgets.
Geo-Fence	A virtual boundary around a customer location; used to validate physical presence before allowing order entry.
Master Sync	The full bi-directional data exchange between the mobile device and the Almashreqs central database, performed at login and on demand.
Offline Queue	A local SQLite table that stores transactions (bookings, logs) when the device lacks internet, awaiting automatic upload.
AI Enhancement	The automated reordering of customer visits in a tour plan to minimize travel distance and time using machine learning heuristics.
Feature Lock	A role-based access control mechanism that hides or disables app features based on the user's position in the 8-tier hierarchy.
End-of-Day Upload	The daily compilation and secure transmission of all field bookings to the central CRM for administrative processing.

12.2 System Requirements and Dependencies

- **Development Setup:** Android Studio Arctic Fox or higher, Flutter SDK 3.0 or higher, Dart 2.17 or greater for the backend code (machine learning), and Python 3.9 or higher for backend code (machine learning).
- **Hardware Specifications:** Android device (4GB RAM, GPS, 4G connection) and development machine (8GB RAM, 50GB of free space).
- **External Services and Accounts:** Google Cloud project with Maps Platform/Firebase integration; Almashreqs API key and VPN access; and GitHub repository with CI/CD secrets set up.
- **Software Libraries and Frameworks:** Firebase Auth, Firestore, FCM SDKs; Google Maps flutter plugin; REST calls via http or dio; Sync in the background using WorkManager; inference on-device using TensorFlow Lite - it's all in one click.

- **Network Requirements:** First time setup and initial Master sync over the Internet, no other network setup is required for offline use other than mobile data.
- **Security Requirements:** The enterprise keystore to sign with; Google Authenticator app on user devices for MFA; TLS 1.3 certificate on Almashreqs API bridge server;